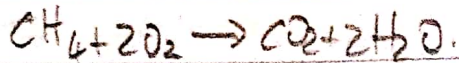


HW 11

(1) Combustion of CH_4



$$\frac{\text{O}_2}{\text{N}_2} \text{ in air} = \frac{21}{79}, \quad \text{N}_2 = 2 \cdot \frac{79}{21} = 7.52 \text{ mol}$$

Energy Balance

$$H_P = H_F + H_R$$

$$\text{For } T = 298\text{K}, H_F = 0$$

$$H_C'_{\text{CH}_4} = 889 \text{ kJ/mol}$$

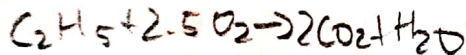
$$C_p(\text{H}_2\text{O}) = 40.2 \text{ J/molK}, \quad C_p(\text{N}_2) = 32.2 \text{ J/molK}, \quad C_p(\text{O}_2) = 32.4 \text{ J/molK}, \quad C_p(\text{CO}_2) = 51.8 \text{ J/molK}$$

$$(T_F - 298)(n \cdot C_p(\text{O}_2) + n C_p(\text{N}_2) + n C_p(\text{H}_2\text{O})) = 889$$

$$(T_F - 298)(51.8 + 7.52 \times 32.2) + (2 \times 40.2) = 889$$

$$T_F = 2663.3 \text{ K}$$

For Acetylene



$$\text{mol N}_2 = 2.5 \cdot \frac{79}{21} = 9.4 \text{ mol N}_2$$

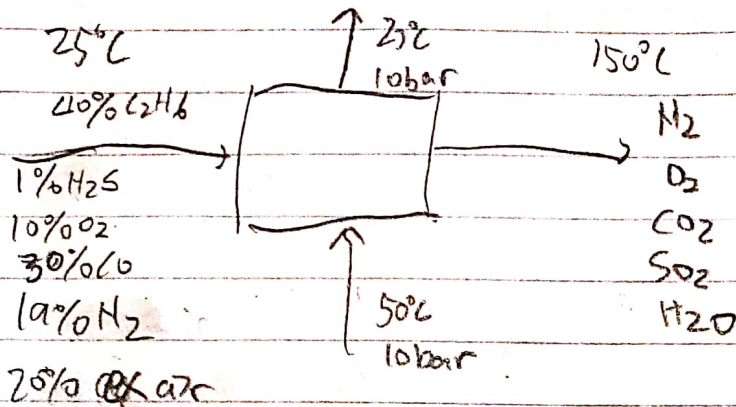
$$H_C'_{\text{C}_2\text{H}_2} = 1300 \text{ kJ/mol}$$

$$(T_F - 298)(2 \times 51.8 + 1 \times 40.2 + 9.4 \times 32.2) = 1300$$

$$T_F = 3209.66 \text{ K}$$

74
85

2. 40 mol% C_2H_6 , 19 mol% H_2 , 1 mol% H_2S , 10 mol% O_2 , 30 mol% CO
 20% excess air $25^\circ C = T_1$, $150^\circ C = T_2$



Basis: 100 mol/s

$$n_{H_2S} = 1$$

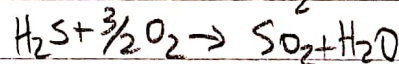
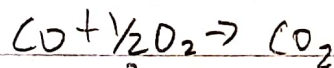
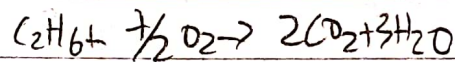
$$n_{O_2} = 10$$

$$n_{H_2} = 19$$

$$n_{C_2H_6} = 40$$

$$n_{CO} = 30$$

Reactions



$$\therefore O_2 \text{ required} = \frac{7}{2}(40) + \frac{1}{2}(30) + \frac{3}{2}(1) - 10 = 146.5 \text{ mol/s}$$

Air: 20% excess

$$n_{O_2, \text{Air}} = 1.2 \times 146.5 = 175.8 \text{ mol/s}$$

$$n_{N_2, \text{Air}} = 175.8 \cdot \frac{79}{21} = 661.3 \text{ mol/s}$$

Balance:

$$C_2H_6: n_{\text{out}, C_2H_6} = 0 = 40 - \dot{F}_1 \quad \dot{F}_1 = 40$$

$$CO: n_{\text{out}, CO} = 0 = 30 - \dot{F}_2 \quad \dot{F}_2 = 30$$

$$H_2S: n_{\text{out}, H_2S} = 0 = 1 - \dot{F}_3 \quad \dot{F}_3 = 1$$

$$CO_2: n_{\text{out}, CO_2} = 2\dot{F}_1 + \dot{F}_2 = 80 + 30 = 110 \text{ mol/s}$$

$$SO_2: n_{\text{out}, SO_2} = \dot{F}_3 = 1 \text{ mol/s}$$

$$H_2: n_{\text{out}, H_2} = n_{i, H_2} + n_{\text{Air}, H_2} = 19 + 661.3 = 680.3 \text{ mol/s}$$

$$H_2O: n_{\text{out}, H_2O} = 3\dot{F}_1 + \dot{F}_3 = 121 \text{ mol/s}$$

$$O_2: n_{\text{out}, O_2} = n_{i, O_2} + n_{\text{Air}, O_2} - \frac{7}{2}\dot{F}_1 - \frac{1}{2}\dot{F}_2 - \frac{3}{2}\dot{F}_3 = 10 + 175.8 - 140 - 15 - 1.5 = 29.3 \text{ mol/s}$$

$$H_{\text{flue}} = \sum \Delta H_c + \sum n \int_{25}^{150} c_p dT$$

$$\Delta H_c: \text{ethane} = -142846$$

$$\text{CO} = -283$$

$$\text{H}_2\text{S} = -51817$$

using table B.17, for c_p , then we get

$$H_{\text{flue}} = -62,400 \text{ kJ/s}$$

$$\Delta H = H_{\text{out}} - H_{\text{in}} = Q = -62400 \text{ kJ/s}$$

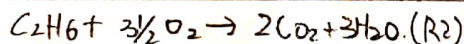
$$m_{\text{steam}} (H_{\text{steam}} - H_{\text{bro}}) = m_{\text{steam}} (2777.1 - 20.1) = 62400 \quad (\text{Steam table})$$

$$m_{\text{steam}} = 24.3 \text{ kg/s}$$

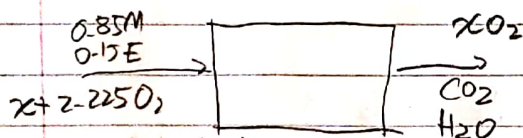
3. 85 mol% methane, ethane (balanced) O_2 , at 25°C

cal basis: 1 mol/s

$$M = 0.85 \text{ mol/s} \quad E = 0.15 \text{ mol/s}$$



$$\text{O}_2: (2 \cdot 0.85) + (3.5 \cdot 0.15) = 2.225 \text{ mol/s} + x$$



$$Q = H_2 - H_1 + H_r \quad \Delta T = 0$$

$$Q = \Delta H_r$$

RT balance.

$$\Delta H_r = H_{\text{prod}} - H_{\text{react}}$$

$$\Delta H_c \text{ CH}_4 = -75.4$$

$$\text{O}_2 = 0$$

$$\text{CO}_2 = -393.5$$

$$\text{H}_2\text{O} = -24.1$$

$$\Delta H_r = -393.5 + 2(-24.1) - (-75.4) = 801.7 \cdot 0.85 = -681.5 \text{ kJ/s}$$

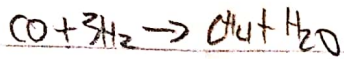
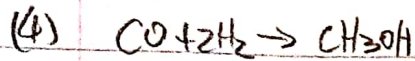
R2.

$$\Delta H_r = 2(-393.5) + (3.5(-24.1)) + (83.82) = -1550.5 \text{ kJ/mol} \cdot 0.15 \text{ mol/s} = -232.5 \text{ kJ/s}$$

$$Q = -681.5 - 232.5 = -914.1 \text{ kJ/s} \times \frac{1 \text{ s}}{\text{mol}} = -914.1 \text{ kJ/mol}$$

(b) would be the same as part 1, since $V = \text{constant}$.

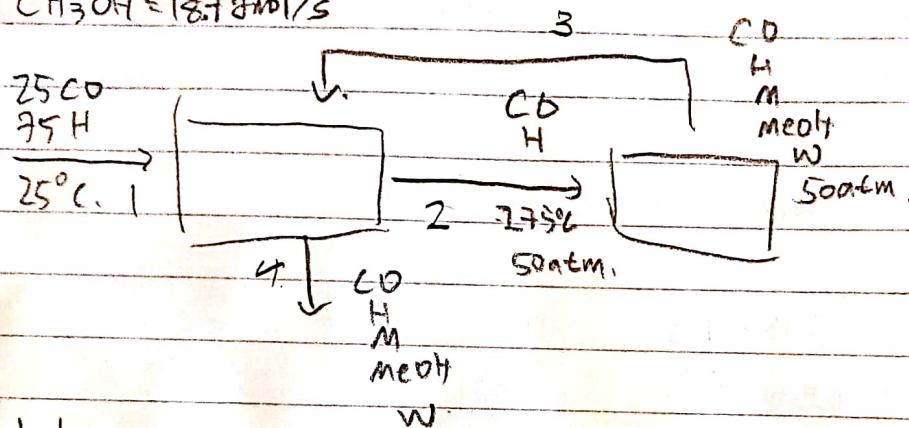
(c) Since we have 100% conversion, $\Delta T = 0$, therefore any excess of O_2 will leave the reactor/bath.



$75 \text{ g mol/s } H_2, 25 \text{ g mol/s } CO \quad T_1 = 25^\circ C \quad T_2 = 275^\circ C$

50 atm, adiabatic. 80% CO $\rightarrow$ Product.

$CH_3OH = 18.7 \text{ g mol/s}$



Mass balance

$$n_{CO,1} = 25$$

$$n_{H_2,1} = 75$$

$$n_{MeOH,3} = 18.7 = \dot{J}_1$$

$$n_{CO,3} - n_{CO,2} = 0.8 \cdot 25 = 20 = \dot{J}_1 + \dot{J}_2 = 18.7 + \dot{J}_2 \Rightarrow \dot{J}_2 = 1.3$$

$$n_{H,3} = 75 - 2(18.7) - 3(1.3) = 33.7$$

$$n_{M,3} = \dot{J}_2 = 1.3 = n_{W,3}$$

Energy Balance

Steady state, adiabatic, $W = 0$

$$H_2 = H_3$$

$$\Delta H_R^\circ = -1(-110.53) + 1(-200.94) = -90.41$$

$$\Delta H_{R,2} = -110.53 - 74.52 - 241.83 - 205.82$$

$$H_3 - H_2 = \sum \Delta H_f^\circ + \sum n_2 \int_{275}^{25} C_p dT + \sum n_3 \int_{25}^{T_3} C_p dT = 0 \quad (B, A)$$

$$T_3 = 1300K$$

Comment Summary

Page 4

1. -7 Not the same.
2. -4 for not having the right solution